GroundingDINO — Findings Report v3

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Overview

This report summarizes exploration of Grounding DINO, an open-vocabulary object detection model, as a potential first-stage component in a deepfake-detection pipeline — specifically, for locating faces or facial regions before a dedicated forensic manipulation-detection model is applied downstream. Version 2 incorporated three additional test conditions requested in supervisor feedback: an expanded occluded-face category, a new small/low-resolution-face category, and an initial (non-forensic) deepfake-image sample.

New in v3: access to RWDF-23 (Real-World Deepfake dataset, DASH Lab) was granted, so the deepfake condition is now tested on genuine in-the-wild deepfake video frames instead of only the Kaggle placeholder (Section 5.8). The Dataset Access Status, Limitations, Challenges, and Next Steps sections are updated to match. Sections 5.1–5.7 are carried over from v2 unchanged, except that the Section 5.5 caveat now points to 5.8.

Model Summary

Grounding DINO addresses a core limitation of conventional object detectors: fixed category lists set during training. Instead of a closed vocabulary, it accepts a natural-language text prompt alongside an image, and grounds objects in the image to phrases in that prompt, allowing detection of arbitrary categories at inference time without retraining.

The model takes two inputs (an image and a text prompt) and fuses them throughout the detection pipeline rather than only at a final classification step, which lets language guide which image regions the model attends to. Its output is a set of bounding boxes, each paired with a confidence score and a phrase drawn from the input prompt describing what was detected.

Two thresholds control the output: the box threshold sets the minimum confidence a candidate box must reach to be kept at all, while the text threshold determines which words in the prompt are matched closely enough to a region to appear in its label. Together they let a user tune the tradeoff between missing real objects and surfacing noisy or duplicate detections.

Setup

Environment

Component	Version / Detail
Python	3.10
PyTorch	2.13.0
Transformers	4.37

Component	Version / Detail
CUDA	N/A (macOS, no NVIDIA GPU)
GPU / Chip	Apple M2
OS	macOS 26.6 (Sections 5.1–5.7); macOS 27.0.1 (Section 5.8)
SciPy	1.15.2 from conda-forge for the macOS 27 runs (see Challenges)

Inference was run in --cpu-only mode rather than via MPS (Metal), since Grounding DINO's MPS path has known compatibility issues (e.g. torch.roll is unsupported on MPS without PYTORCH_ENABLE_MPS_FALLBACK=1, and cumsum lacks int64 support, affecting accuracy), and CPU mode benchmarks comparably fast in community tests. The official Swin-T checkpoint was downloaded and basic inference was validated via the repository's official demo script before building a custom batch pipeline.

Methodology

A custom batch-inference script runs Grounding DINO across a folder of images with configurable prompt and threshold values via CLI arguments, saving annotated images and a predictions.json file containing per-image bounding box coordinates, matched phrases, confidence scores, and inference time. Across the original runs, inference averaged 3.37 seconds per image in CPU-only mode on the Apple M2.

Test Image Set (v3)

Following feedback that the occluded category was too thin (2 images) to draw confident conclusions from, and that a small/low-resolution face condition was missing entirely, the test set was expanded in v2. The multiple-faces category also gained one new image that round (see Section 5.2). v3 adds 26 RWDF-23 frames.

Category	Count	Description
Single frontal face	10	LFW via Hugging Face mirror logasja/lfw
Multiple faces	5 (was 4)	+1 new: one additional dense crowd scene (Unsplash)
Profile / angled face	5	Stock photos (Unsplash / Pexels)
Occluded / blurred face	7 (was 2)	+5 new: masks, hands, hair, sunglasses+scarf, partial crops (Unsplash/Pexels)
Small / low-resolution face	6 (new in v2)	Thumbnail-resolution downscales (32–90px long edge) of already-curated photos
Deepfake sample (placeholder)	12 (new in v2)	Kaggle manjilkarki/deepfake-and-real-images, Test/Fake split — non-forensic substitute
Deepfake — RWDF-23	26 (new in v3)	Frames from 20 in-the-wild deepfake videos (8 TikTok, 5 Reddit, 7 Bilibili) listed in RWDF-23 (Cho et al., 2023)

small_lowres images are thumbnail-resolution downscaled copies of already-curated photos from single/profile/occluded (long edge 32–90px), isolating resolution as the sole variable rather than introducing new subjects/poses at the same time. The Kaggle deepfake sample is an openly-downloadable, non-forensic set that served as a placeholder until RWDF-23 access was granted; it is kept for comparison in Section 5.8.

Results

5.1 Prompt Experiments (Original Categories, Recap)

• “face . eyes . mouth” consistently produced more boxes than “human face” across every category (e.g. single: 3–11 boxes/image vs. 1–4 boxes/image) — expected, since the .-separated prompt creates three distinct grounding targets (face, eyes, mouth) rather than one box per face.

• “real face” and “manipulated face” tracked closely with “human face” in nearly all cases, indicating the model is not meaningfully responding to the “real”/“manipulated” qualifier — it continues to primarily match on “face.”

• The profile and (original 2-image) occluded categories were largely stable at 1 box/image across face-related prompts.

• “person” produced comparable box counts to “human face” on profile/occluded samples, but diverged on multiple-faces and single-face categories — suggesting “person” and “human face” generalize differently to crowds vs. individuals.

5.2 Multiple Category (Expanded, 4 → 5 images)

One new image was added to the multiple-faces category — a large, genuinely crowded stadium/rally photo (hundreds of people, individual faces roughly 30–40px) — bringing the category from 4 to 5 images. At default thresholds (box=0.35/text=0.25), this new image returned zero detections for four of the five standard prompts:

Prompt	Boxes on new crowd photo
human face	0
person	0
face . eyes . mouth	0
real face	0
manipulated face	1 (score 0.44 — likely noise)

For comparison, the original multiple-faces outlier from the first round (nicholas-green, a smaller group) still returns 18 boxes on “human face” at the same threshold (see Section 5.6). This is a different failure mode than the duplicate-box behavior documented for that image — rather than over-detecting a crowded scene, the model misses it entirely once faces get small/dense enough. As the threshold sweep in Section 5.7 shows, this isn't recoverable by threshold tuning either.



Figure D. New multiple-faces image at the most permissive threshold tested (box=0.25/text=0.20) — 2 low-confidence boxes (0.296, 0.264) are visible if you look closely; at default and higher thresholds this image returns 0 boxes despite hundreds of visible faces.

5.3 Occluded Category (Expanded, 2 → 7 images)

With six new images added — covering masks, hands covering the face, hair covering the face, and sunglasses+scarf combinations — detection at default threshold (box=0.35/text=0.25, prompt “human face”) was highly consistent: 6 of 7 images produced exactly 1 box. The one exception was a hair-covering-face image, which produced 2 overlapping boxes at different confidences (0.60 and 0.39) — suggesting the model can produce a duplicate detection when hair partially bisects, rather than fully occludes, the visible face region.

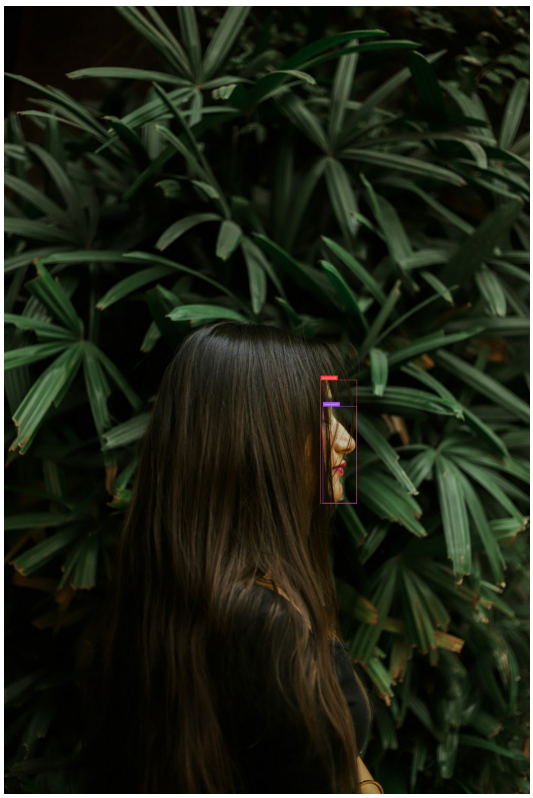


Figure A. Occluded category example — hair partially covering the face, prompt “human face”, default thresholds. This image produced 2 overlapping boxes (0.60, 0.39) rather than the single box typical of the rest of the expanded category.

5.4 Small / Low-Resolution Category (New, 6 images)

Downscaling existing single-subject photos (single, profile, occluded) to thumbnail resolution (32–56px long edge) did not break detection: all three produced exactly 1 box, at confidence scores (0.57–0.82) comparable to their full-resolution originals. Downscaled multi-subject group photos (90px long edge) still detected multiple faces (4–5 boxes at default threshold) but at reduced count and confidence relative to the full-resolution source — partial rather than complete degradation under combined multi-subject + low-resolution conditions.



Figure B. small_lowres example, shown enlarged from its native 90×60px — a downscaled group photo, prompt “human face”, default thresholds. 5 of the original group's faces were still detected at this resolution.

5.5 Deepfake-Sample Prompt Results (Exploratory)

The five standard prompts were run against the 12-image Kaggle placeholder sample (Test/Fake split, manjilkarki/deepfake-and-real-images) at default thresholds:

Prompt	Avg boxes/img	Total boxes (12 img)	Notes
human face	1.33	16	baseline
person	1.58	19	slightly more than “human face”
face . eyes . mouth	4.58	55	expected — separate targets, matches original pattern
real face	1.25	15	tracks closely with “human face”
manipulated face	1.17	14	tracks closely with “human face”; 1 zero-detection (only one across all 5 prompts)

“Real face” (1.25) and “manipulated face” (1.17) both sit in the same range as plain “human face” (1.33) — not meaningfully lower or differently distributed. This replicates the original-round finding on ordinary photos, now on a deepfake-labeled (if non-forensic) sample: the model is not detecting anything forensically different between the qualifiers.



Figure C. Deepfake-sample example (Kaggle Test/Fake), prompt “human face”, default thresholds, score 0.83.

Caveat: this is an openly-downloadable Kaggle image set, not a forensic deepfake-detection benchmark. Results here are exploratory only. Section 5.8 repeats the comparison on genuine deepfake frames from RWDF-23.

5.6 Threshold Sensitivity — Original Categories (Recap)

Three representative images (one each from multiple, single, and profile, selected for high or borderline box counts) were re-run at three box/text threshold combinations:

Image (category)	0.25/0.20	0.35/0.25	0.45/0.30	Interpretation
lfw_02 (single)	4 boxes	4 boxes, identical	1 box	Extra boxes at low threshold are genuine background faces, not duplicates.
nicholas-green (multiple)	26 boxes	18 boxes	1 box	Heavy duplication at low threshold; highest threshold undercounts a genuinely crowded scene.
pexels-moh-adbelghaffar (profile/occluded)	3 boxes	1 box	0 boxes	Highest threshold misses the face entirely.

5.7 Threshold Sensitivity — Multiple, Occluded, and Small/Low-Resolution

The same three threshold combinations were run against the expanded/new categories:

Image (category)	0.25/0.20	0.35/0.25	0.45/0.30	Interpretation
neelakshi-singh (multiple, new dense crowd)	2 boxes [0.296, 0.264]	0 boxes	0 boxes	Even the most permissive threshold recovers only 2 very-low-confidence boxes out of hundreds of visible faces. Not a threshold-tuning problem — the scene is effectively invisible to the model at every setting tested.
yogesh-rahamatkar (occluded)	2 boxes [0.386, 0.399]	2 boxes, identical	0 boxes	Only occluded image sensitive to threshold — both detections score below 0.45. All other occluded images stayed at 1 box across all three thresholds.
thumb_multiple_02 (small_lowres)	8 boxes	5 boxes	1 box	Sharpest decline of any small_lowres image — a downscaled group photo loses most detections well before the highest threshold.
thumb_single_lfw03 (small_lowres)	2 boxes	1 box	1 box	Extra box at lowest threshold is a low-confidence duplicate that disappears by 0.35/0.25; otherwise stable.

Two distinct patterns emerge. The expanded occluded category and the single/profile-derived small_lowres images behave like the original categories: threshold mainly trades off duplicate/spurious boxes against missed genuine ones, and 0.35/0.25 remains a reasonable general-purpose default. But the dense-crowd case doesn't respond to threshold tuning the same way — lowering the threshold recovers duplicate boxes on a scene the model already detects (as with the original nicholas-green outlier, Section 5.6), but does almost nothing for a scene it fundamentally failed to detect in the first place. Threshold sweeping is not a fix for the crowd-density failure mode documented in Section 5.2.

5.8 RWDF-23 Real-World Deepfake Frames (New in v3)

Sample. RWDF-23 (Cho et al., 2023) is distributed as a metadata spreadsheet of links to deepfake videos posted on social platforms (1,998 unique links: YouTube 1,190, TikTok 598, Reddit 111, Bilibili 100). A seeded, platform-stratified sample of 20 links was downloaded with yt-dlp (720p max), and further untried links were drawn (cycling TikTok, Reddit, Bilibili, YouTube) until 20 videos had downloaded. That took 40 attempts:

Platform	Tried	Downloaded	Failure reasons
YouTube	12	0	9 blocked by a sign-in / bot check (account cookies not used); 3 videos removed
TikTok	12	8	4 posts blocked for this IP address (region restriction)
Reddit	8	5	3 posts deleted (HTTP 404)
Bilibili	8	7	1 downloader extractor error
Total	40	20	20 failed: 6 removed/deleted, 14 access-controlled or tool errors

From the 20 videos, 6 evenly spaced frames each were extracted and 26 frames with clearly visible faces were kept (1–2 per video), chosen by eye rather than by running Grounding DINO first so the set is not pre-filtered toward easy cases. The frames cover tight face-swap close-ups, two- to nine-face scenes, split-screen and stacked-panel edits, a TV news broadcast with on-screen graphics, a dim film still, and four frames in which someone holds up a tablet, phone, or printed photo showing a face. All 5 standard prompts were run at default thresholds (box=0.35/text=0.25). RWDF-23 frames are licensed data and are not reproduced in this report.

Prompt	RWDF-23 avg boxes/img (26)	Kaggle avg boxes/img (12)	RWDF-23 mean score
human face	1.92	1.33	0.678
person	2.08	1.58	0.678
face . eyes . mouth	6.42	4.58	0.526
real face	1.92	1.25	0.713
manipulated face	1.81	1.17	0.699

RWDF-23 averages are higher than Kaggle's mainly because half of the RWDF-23 frames contain more than one face.

Do “real face” / “manipulated face” still just track “human face” on genuine deepfakes? Yes. To go beyond average box counts, each “real face” and “manipulated face” box was matched to its highest-overlap “human face” box on the same image:

Comparison	RWDF-23 (26 img)	Kaggle (5.5)
Images with the same box count as “human face” — real / manipulated	24/26 · 24/26	11/12 · 10/12
Matched boxes with IoU > 0.9 — real / manipulated	48/50 · 40/47	15/15 · 13/14
Mean score change vs. “human face” — real / manipulated	+0.025 · +0.000	+0.024 · –0.045
Same box: “manipulated face” scores above “real face”	7 of 46 (mean –0.033)	0 of 14 (mean –0.074)

The qualifiers select the same regions as “human face”, with nearly identical scores. The 7 “manipulated face” boxes below IoU 0.9 still overlap at 0.71–0.90 (the same face with slightly different box edges), and the 2 unmatched “real face” boxes are faces that “human face” missed. Every frame comes from a deepfake video, yet “manipulated face” scores slightly *below* “real face” on the same box in 39 of 46 cases, the same direction as on the Kaggle sample. That is a constant bias from the prompt wording, not a response to manipulation. The Kaggle result from Section 5.5 holds on forensic-quality, in-the-wild content.

Localization holds up on real deepfake frames. By visual inspection, “human face” found 44 of the 46 faces physically present in the scenes. All 13 single-face frames returned exactly 1 box (scores 0.81–0.89). Face-swap blending, app watermarks (e.g. “Face swap made with Impressions app”), burned-in captions, news graphics, and split-screen/stacked edits did not disrupt detection. The two misses were a face turned away and mostly covered by hair and a child with her head bowed, both pose/occlusion cases rather than anything specific to manipulation.

New failure modes on in-the-wild video:

- **Faces on screens and prints are localized as faces, sometimes above the real face.** In 4 of the 26 frames someone holds up a tablet, phone, or printed photo showing a face (an illustration, a drawing, a photo, and two faces on a phone screen). All 5 of these faces were boxed by every face prompt (scores 0.37–0.83). In one frame the face on the tablet (0.82) outscored the real person holding it (0.70). In the two-stage pipeline these crops would go to the forensic classifier and likely be flagged as “fake”, a false positive introduced by stage 1. Holding up the “source” face appeared in 4 of the 20 sampled videos.

- **Prompt wording shifts recall at the threshold margin.** On a three-face frame, “real face” recovered the bowed child's face (0.38) that “human face” and “manipulated face” missed; on a nine-face stacked frame, “manipulated face” found 7 faces vs. 9; on a two-person frame, “human face” produced a duplicate box (0.61 + 0.37) that the other two did not. The qualifier carries no authenticity signal, but it moves borderline faces across the 0.35 cutoff, so face counts can differ by prompt.

- **Lower confidence in multi-face frames.** “human face” averaged 0.62 on multi-face frames (range 0.36–0.82) versus 0.86 on single-face frames, and the faces other than the top-scoring one averaged 0.56. These are the first faces lost at higher thresholds, consistent with Sections 5.6–5.7.
- **“eyes” splits three ways.** On a clear frontal face “face . eyes . mouth” returns left eye, right eye, and a combined both-eyes box (exactly 3 “eyes” boxes on 12 of 13 single-face frames), which inflates box counts.

Caveats: 26 frames from 20 videos is still a small sample with no YouTube content, so results are directional. RWDF-23 labels are per video, so it is not known which face in a multi-face frame was swapped; the analysis above is about localization and prompt behavior, not per-face authenticity.

Dataset Access Status (deepfake condition)

Dataset	Access	Status
FaceForensics++	Request form	Pending as of v2 (no update since Aug 2026)
RWDF-23 (DASH Lab)	Google Form	**Granted (Oct 2026)** — used in Section 5.8
FakeAVCeleb (DASH Lab)	Google Form + license agreement	Requested as of v2 (no update since Aug 2026)

RWDF-23 replaces the Kaggle sample as the primary deepfake condition. The Kaggle sample (Section 5.5) is kept only as a point of comparison. RWDF-23 data (metadata, video links, downloaded videos, frames) is kept local and is not redistributed in the project repository.

Limitations

- **Detection can fail completely, not just duplicate/over-detect, once a scene is dense/small enough,** and this failure does not resolve with threshold tuning (Sections 5.2, 5.7). This is a more significant limitation than the duplicate-box issue alone for any pipeline relying on this as a first-stage face locator.
- **RWDF-23 coverage is small and skewed.** 26 frames from 20 videos, with no YouTube content (about 60% of the dataset) because all 12 YouTube links tried were blocked or removed. Labels are per video, not per face.
- **Faces on screens and prints pass stage 1.** Faces on held-up devices or photos are localized like real ones (Section 5.8) and would reach the forensic classifier without an additional filter.
- **Occluded (7) and multiple (5) category sizes are improved but still modest** — findings remain directional, not statistically robust.
- **No single threshold setting is universally correct** across categories or resolutions; a production pipeline would need either a per-scene-type threshold policy or an alternative to a single global cutoff, such as adaptive thresholding.
- **Duplicate box proposals** at lower thresholds continue to appear (observed on crowded, occluded hair-covering, and now one RWDF-23 frame) — raw detection output requires

post-processing (e.g. non-max suppression tuning) before use as input to a downstream forensic model.

• **Small/low-resolution degradation is partial, not binary,** for the conditions tested here — single-subject detection was robust down to ~32px, while multi-subject detection degraded gradually rather than failing outright.

• **Box counts and visual inspection, not ground truth.** No face-box annotations were used; “missed” and “duplicate” come from visual review.

Challenges

• Function signature mismatches between the CLI demo script and the underlying groundingdino.util.inference API (load_model and predict each have independent device parameters with different defaults), requiring direct inspection of the source rather than assuming consistent behavior.

• A transformers version incompatibility (AttributeError: 'BertModel' object has no attribute 'get_head_mask') required pinning to transformers==4.37.

• The threshold-sweep script initially failed silently: a bash idiom (set -- $combo) used to split a “box text” pair does not word-split the same way in zsh (macOS's default shell) as in bash, so all sweep runs errored out with an invalid-float-value argparse error before writing any output. Fixed by switching to shell-portable parameter-expansion splitting (${combo%%:*} / ${combo##*:}).

• **New in v3:** after upgrading to macOS 27, the PyPI SciPy 1.15.3 wheel failed to load (“'__DATA/__thread_bss' has a zero-fill section type, but offset field is not zero”), breaking every inference run. Fixed by installing SciPy 1.15.2 from conda-forge.

• **New in v3:** retrieving RWDF-23 videos. An outdated yt-dlp failed on every platform; upgrading it fixed Reddit and Bilibili, but YouTube now requires a signed-in session and some TikTok posts are region-blocked, so reaching 20 working videos took 40 attempts (Section 5.8).

Next Steps

• Enlarge the RWDF-23 sample, including YouTube via an authenticated session, and add per-face labels for which face in each frame was swapped.

• Pass --save_crops output from RWDF-23 frames to a pretrained deepfake classifier to measure end-to-end (stage 1 + stage 2) accuracy.

• Filter faces shown on held-up screens or prints between stage 1 and stage 2.

• Investigate the crowd-density detection failure further (Section 5.2) — e.g. whether tiling the image before inference, or running at a higher input resolution, recovers detections that a single global threshold cannot.

• Investigate non-max suppression or other post-processing to address duplicate box proposals, now observed in crowded, occluded (hair-covering), and RWDF-23 cases.

• Consider a per-category or adaptive threshold policy rather than a single global default, given how differently occluded, small/low-resolution, and multi-subject images respond to threshold changes.

References

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